Unit: 6 Unit 8: Quadratics **Name:** ________________
Topic: Definition of Quadratics and Standard Form **Date:** __________

Advanced (Habanero)

Q1. What is the standard form of a quadratic function?
(A) $ax^2 + bx + c$
(B) $ax + bx^2 + c$
(C) $ax^2 - bx + c$
(D) $ax^3 + bx^2 + c$
(E) $ax^2 + b$

Q2. Which of the following is a quadratic function in standard form?
(A) $x^2 + 3x - 2$
(B) $x^3 - 2x^2 + x$
(C) $x^2 - 4$
(D) $2x^2 + 5x$
(E) $x^4 - 2x^2 + 1$

Q3. What is the value of a in the quadratic function $2x^2+3x-1$?
(A) 1
(B) 2
(C) 3
(D) -1
(E) -2

Q4. Which of the following quadratic functions has a vertex at $(0, 0)$?
(A) $x^2 + 2x + 1$
(B) $x^2 - 2x - 1$
(C) x^2
(D) $-x^2$
(E) $x^2 + 1$

Q5. What is the axis of symmetry of the quadratic function $x^2 + 2x + 1$?
(A) $x = -1$
(B) $x = 1$
(C) $x = 0$
(D) $x = -2$
(E) $x = 2$

Q6. Which of the following is a key feature of the quadratic function $x^2 + 2x + 1$?
(A) Vertex at $(0, 0)$
(B) Axis of symmetry at $x = 0$
(C) VERTEX at $(-1, 0)$
(D) Axis of symmetry at $x = -1$
(E) Vertex at $(1, 0)$

Q7. What is the y-intercept of the quadratic function x^2+2x+1?
(A) $(0, 0)$
(B) $(0, 1)$
(C) $(0, -1)$
(D) $(1, 0)$
(E) $(-1, 0)$

Q8. Which of the following quadratic functions has a y-intercept at $(0, 2)$?
(A) $x^2 + 2x + 1$
(B) $x^2 + 2x + 2$
(C) $x^2 - 2x + 2$
(D) $x^2 - 2x - 2$
(E) $x^2 + 2$

Open Ended Questions (FRQ)

Q9. Describe the key features of the quadratic function x^2-4x+3. Be sure to include the vertex, axis of symmetry, and y-intercept.

Q10. Write a quadratic function in standard form with a vertex at $(-2, 3)$ and a y-intercept at $(0, 7)$. Show all work and explain your reasoning.

Chef's Tips

- Tip 1: Make sure to check for any calculations or work shown for free response questions
- Tip 2: Consider using a graphing calculator to verify student answers for quadratic functions
- Tip 3: Encourage students to use proper notation and formatting when writing quadratic functions